1. What is a "wait-for graph" used for in deadlock detection?

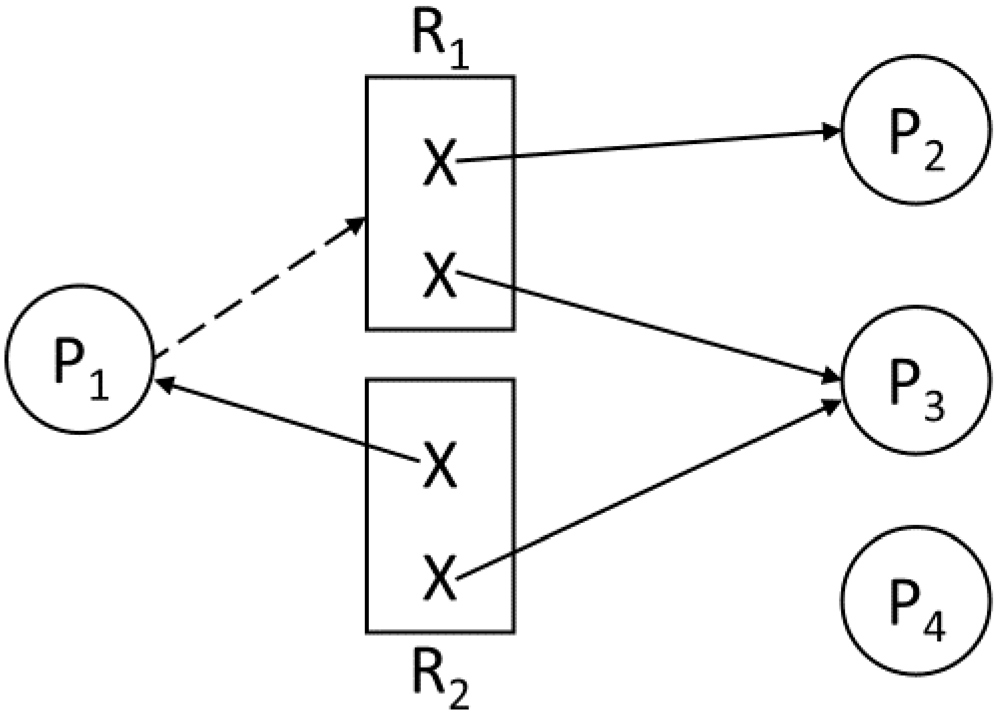
A. To prevent deadlock by prioritizing processes.

B. To represent relationships between processes and resources.

C. To check for cycles indicating potential deadlocks.

D. To allocate resources to processes dynamically.

2. Based on the resource allocation graph shown in **Figure 1**, select the best statement that represents the graph.



A. P1 requested for an instance of R1 and was assigned an instance of R2.

B. P1 holds an instance of resource R1 and requested for an instance of R2.

C. P3 requested for an instance of R1 and R2.

D. The system is in an unsafe state.

3. The __________ condition in deadlock detection occurs when a process is indefinitely waiting for a resource that another process holds.

A. Mutual Exclusion

B. Circular Wait

C. Hold and Wait

D. Preemption

4. Which conditions can resolve deadlocks through resource preemption?

I. Selecting the victim process to preempt resources.

II. Forcing processes to release resources voluntarily.

III Rolling back processes to a previous safe state.

IV. Prioritizing the most recently executed processes.

A. II and IV

B. II and III

C. I and IV

D. I and III

5. In memory management, how does dynamic partitioning reduce fragmentation?

A. By using fixed-sized memory blocks.

B. By merging adjacent free memory blocks.

C. By swapping processes in and out of memory.

D. By assigning equal partitions to all processes.

6. Given a page size of bytes and a logical address of 16MB, how many bits represent the page offset?

A. 8 bits.

B. 16 bits.

C. 24 bits.

D. 32 bits.

7. The ______ bit in a page table entry indicates whether a page is present in memory or not.

A. Valid-invalid

B. Access control

C. Read-write

D. Relocation

8. In a system that uses demand paging, which of the following actions does the operating system NOT perform when handling page faults?

A. Load the required page from disk into a free frame.

B. Update the page table to reflect the new page's location.

C. Select a page to evict if there are no free frames.

D. Execute the process that caused the page fault immediately after loading the page.

9. Select the best statement(s) that best represents page replacement.

1. Page replacement occurs when a page fault occurs.

2. Page replacement occurs when there are no free frames available.

3. The performance metric of a page replacement algorithm is based on the minimum number of page faults.

4. A page can only be brought into memory once.

A. I and II

B. I, II, and III

C. I , III and IV

D. II, III and IV

10. When virtual memory is used, which of the following are **true** regarding its functionality?

1. It allows processes to exceed the available physical memory.

2. It uses a page table to map virtual addresses to physical addresses.

3. Virtual memory increases system performance by reducing the need for secondary storage.

4. Pages in virtual memory are fixed-size blocks, and the system employs paging to manage them efficiently.

A. I, II and IV

B. II, III and IV

C. I, III and IV

D. I, II and III

11. The process of moving a page between disk and memory to handle a page fault is called _______, and if this process occurs too frequent, it leads to _______.

A. paging, thrashing

B. paging, fragmentation

C. swapping, thrashing

D. swapping, fragmentation

12. Which of the following is an advantage of contiguous allocation for file storage?*

A. It eliminates fragmentation by using a linked list of blocks.

B. It provides fast access to files, but suffers from external fragmentation.

C. It allows for dynamic resizing of files during runtime.

D. It is efficient for handling large files that require random access.

13. In magnetic tape storage, data is accessed in a __________ manner, requiring the tape to be read from the beginning to the desired location, making it less suitable for __________ access.

A. random, sequential

B. sequential, random

C. block-based, random

D. sequential, disk-based

14. In a file system, seek time is the time taken for the disk's read/write head to move from its current position to the __________ track, while access latency refers to the time it takes for the disk to rotate the correct __________ under the head.

A. outermost, sector

B. innermost, track

C. correct, block

D. desired, sector

15. What is the primary disadvantage of using linked allocation in a file-system?*

A. The need for extra memory to store pointers in each block.

B. External fragmentation due to scattered file blocks.

C. Difficulty in accessing a file's data block sequentially.

D. Inefficient handling of files requiring contiguous storage.

16. In disk scheduling, the difference between innermost and outermost tracks refers to ____________.

A. The time it takes to access the innermost and outermost sectors.

B. The total number of tracks available on the disk.

C. The physical location of the heads, with outer tracks closer to the spindle.

D. The positioning time for accessing the innermost and outermost disk heads.

17. In a file system that uses a bit map to manage free space, a bit is set to ___ when a block is free, and to ___ when it is allocated.*

A. 1, 0

B. 0, 1

C. 1, 1

D. 0, 0

18. Consider the performance metrics for magnetic disks. Which of the following statements are correct?

1. The seek time is the time it takes for the disk arm to move to the correct track.

2. Access latency is the time taken for the correct sector to rotate under the read/write head.

3. Transfer rate refers to the time taken for the disk to spin from the innermost to the outermost track.

4.Head crash occurs when the disk's read/write head physically contacts the disk surface, causing potential damage to both the head and the data.

A. I, II, IV

B. II, III, IV

C. I, II, III

D. I, III, IV

19. Which of the following statements about RAID are **true**?

1. RAID 1 uses mirroring to replicate data across multiple disks, offering redundancy but no performance improvement.

2. RAID 5 offers better performance than RAID 1 by using striping with distributed parity, allowing for data recovery in case of a single disk failure.

3. RAID 6 provides double parity, allowing the system to tolerate two simultaneous disk failures.

4. RAID 0 offers no redundancy but improves performance by striping data across multiple disks for faster read and write operations.

*

A. I, II and III

B. II, III and IV

C. I, II and IV

D. I, II, III and IV

20. Which of the following statements about magnetic disks are **true**?

1. Magnetic disks have components such as platters, heads, and arms to perform read and write operations.

2. The speed of a magnetic disk is determined by factors such as transfer rate, positioning time, and random-access time.

3. A head crash occurs when the disk's read/write head touches the spinning platter, causing permanent data loss.

4. Performance metrics for magnetic disks include seek time, access latency, and transfer rate, all of which impact overall disk throughput.

*

A. I, II and III

B. II, III and IV

C. I, III and IV

D. I, II and IV